# Derivation of the dual problem

$f_0 \quad \frac{1}{2}|\underline{w}|^2$

$f_\alpha \quad -\{y_T^{(\alpha)}(\underline{w}^T \underline{x}^{(\alpha)} + b) - 1\} \leq 0$

Lagrangian

$\lambda_\alpha \geq 0$

$$L = \frac{1}{2}|\underline{w}|^2 - \sum_{\alpha=1}^{P} \lambda_\alpha \{y_T^{(\alpha)}(\underline{w}^T \underline{x}^{(\alpha)} + b) - 1\}$$

$$\frac{\partial L}{\partial w_\ell} = w_\ell - \sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)} x_\ell^{(\alpha)} \overset{!}{=} 0 \rightsquigarrow w_\ell = \sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)} x_\ell^{(\alpha)}$$

$$\frac{\partial L}{\partial b} = -\sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)} \overset{!}{=} 0 \rightsquigarrow \sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$\left.\begin{aligned} L = \frac{1}{2} &\sum_{\alpha,\beta=1}^{P} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} (\underline{x}^{(\alpha)})^T \underline{x}^{(\beta)} \\ &- \sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)} \sum_{\beta=1}^{P} \lambda_\beta y_T^{(\beta)} (\underline{x}^{(\beta)})^T \underline{x}^{(\alpha)} \end{aligned}\right\} -\frac{1}{2} \; ?$$

$$- \left(\sum_{\alpha=1}^{P} \lambda_\alpha y_T^{(\alpha)}\right) b \leftarrow 0$$

$$+ \sum_{\alpha=1}^{P} \lambda_\alpha \leftarrow$$

$$L = -\frac{1}{2} \sum_{\alpha,\beta=1}^{P} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} (\underline{x}^{(\alpha)})^T \underline{x}^{(\beta)} + \sum_{\alpha=1}^{P} \lambda_\alpha$$